Probe Complexity of Local Model Statistics in #SAT

Fabrizio De Palma
Independent Researcher

Abstract

We study how much information about the local distribution of models of a CNF formula can be extracted through finite counting probes. Fix a tuple of variables $X = (x_1, \dots, x_r)$. Each satisfying assignment induces a pattern in $\{0,1\}^r$, and hence every formula determines a local model distribution on these patterns. A probe is a selected CNF extension whose count effect depends only on this local distribution.

We introduce the notion of **abstract probe complexity**, defined as the minimum number of probes required to determine a given local model statistic. For statistics induced by a partition of $\{0,1\}^r$ into m blocks, we prove that the exact abstract probe complexity is $m - 1$. This yields a complete characterization of the information dimension of partition statistics in $\#SAT$. As consequences, we obtain exact bounds for several natural quantities, including single-variable bias, equality and disequality statistics, parity, and complete local model tomography, the latter requiring exactly $2^r - 1$ probes.

We then study a different task, namely **universal count alteration**: the minimum number of probes required to guarantee a strict increase of the model count for every satisfiable formula. Here we prove an exact threshold: one non-tautological probe never suffices, while two probes always suffice for every arity $r \geq 2$.

The second part of the paper introduces **realizable probe complexity**, in which probes must be implemented by CNF formulas subject to structural constraints. In particular, we study old-width bounds, i.e. bounds on the number of original variables occurring in each probe clause. For parity on r variables, the abstract probe complexity is 1, but no single exact parity probe exists under the restriction $\mathrm{ow} < r$, while two probes always suffice. This provides a first explicit gap between abstract and realizable probing.

We conclude with open problems on the structural limits of probe realizability, including a conjectured exponential trade-off between clause support size and realizable probe complexity for parity.

Keywords: #SAT, Model counting, Boolean satisfiability, CNF formulas, Probe complexity, Local model statistics, Local distributions, Counting probes, Partition statistics, Parity constraints, Combinatorial counting, Computational complexity, Theoretical computer science.

---

1. Introduction

The model counting problem $\#SAT$ asks for the number of satisfying assignments of a CNF formula. It is a fundamental counting problem in theoretical computer science and a central object in automated reasoning, probabilistic inference, verification, and knowledge compilation (Darwiche & Marquis, 2002; Biere, Heule, van Maaren, Walsh, 2021). Most existing work studies $\#SAT$ from an algorithmic or complexity-theoretic viewpoint, focusing on exact computation, approximation, tractable fragments, or structural decompositions (Huang & Darwiche, 2005; Capelli, Durand, Mengel, 2014; Capelli, 2017).

In this paper we study a different question. Rather than asking how to compute $\#F$ itself, we ask what information about the **local distribution of models** of a CNF formula F can be extracted through a finite number of controlled counting experiments.

Fix a tuple of variables

$$X = (x_1, \dots, x_r).$$

Each satisfying assignment of F induces a pattern in $\{0,1\}^r$, and hence F determines a local model distribution on these 2^r patterns. A **local model statistic** is any quantity depending only on this distribution. Natural examples include the bias of a variable, equality and disequality statistics on a pair of variables, parity on a tuple of variables, and the full local model distribution itself.

We study these quantities through selected CNF extensions, called **probes**, whose count effect depends only on the local model distribution under consideration. Concretely, if

$$F' = F \wedge H$$

is a CNF extension obtained by adjoining an auxiliary CNF formula H, the associated probe observation is

$$\#(F \wedge H) - \#F.$$

The basic requirement is that this quantity be determined solely by the local model distribution on the chosen tuple X, uniformly over all formulas containing X.

This leads to the notion of **abstract probe complexity**: the minimum number of probes required to determine a given local model statistic. Our first main result gives a complete characterization of this quantity for statistics induced by partitions of the local pattern space.

Partition Probe Theorem.

Let

$$\Pi = \{B_1, \dots, B_m\}$$

be a partition of $\{0,1\}^r$. For a CNF formula F, let M_j denote the number of models of F whose pattern on X lies in B_j. Then the partition statistic

$$(M_1, \dots, M_m)$$

has abstract probe complexity exactly

$$m - 1.$$

Thus the probing cost of a partition statistic coincides exactly with its number of independent degrees of freedom. Several consequences follow immediately. In particular:

- the bias of a variable has abstract probe complexity 1;
- equality and disequality statistics on two variables have abstract probe complexity 1;
- parity on r variables has abstract probe complexity 1;
- complete local model tomography on r variables has abstract probe complexity $2^r - 1$.

The last statement shows that the full local model distribution can be reconstructed through a finite family of counting probes, while much coarser statistics require only constant probing cost.

We then study a different task, namely **universal count alteration**. Here the goal is not to reconstruct a statistic, but to guarantee a strict increase of the model count for every satisfiable formula. We prove an exact threshold: one non-tautological probe never suffices, while two probes always suffice for every arity $r \geq 2$. This yields a sharp separation between universal count alteration and exact local reconstruction.

The second part of the paper concerns **realizable probe complexity**. The abstract theory above treats probes as ideal observables. In contrast, realizable probes must be implemented by CNF formulas subject to explicit syntactic constraints. To capture this, we introduce structural classes of probes and study the corresponding minimum probing cost under realizability requirements.

Our main structural example is parity. Although parity on r variables has abstract probe complexity 1, we show that this equality fails under natural CNF restrictions. More precisely, if every clause of a probe contains fewer than r original variables, then no single exact parity probe exists; nevertheless, two such probes always suffice. This yields a first explicit gap between abstract and realizable probing.

Taken together, these results suggest that local probing in $\#SAT$ admits two complementary theories. The first is abstract and information-theoretic: it measures the intrinsic dimension of local model statistics. The second is structural: it measures the cost of realizing probes under syntactic CNF constraints. Understanding the relation between these two levels, and in particular the size and nature of the resulting realizability gaps, is the main structural question opened by the present work.

1.1 Organization of the paper

The paper is divided into two parts.

The first part develops the abstract theory of local probing in $\#SAT$. Section 2 introduces local model distributions, local statistics, and probes. Section 3 defines abstract probe complexity and proves the Partition Probe Theorem. Section 4 derives consequences for bias, equality and disequality, parity, and complete local model tomography. Section 5 studies universal count alteration and proves that, for every arity $r \geq 2$, one probe never suffices while two probes always suffice.

The second part studies realizability under structural CNF constraints. Section 6 introduces realizable probe complexity and structural probe classes, including connectedness and old-width. It also proves that parity exhibits a strict gap between abstract and realizable probing under the restriction $\mathrm{ow} < r$. Section 7 sharpens this result by establishing the exact value

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}<r}}(\phi_{\oplus,r}) = 2$$

and by deriving a general upper bound for smaller old-width. Section 8 formulates open problems and conjectures concerning the realizable complexity of parity and of general partition statistics. Section 9 concludes.

1.2 Table of contents

Section 2 – Local model distributions, local statistics, probes

Section 3 - Abstract probe complexity; Partition Probe Theorem

Section 4 - Bias, equality/disequality, parity, local tomography

Section 5 - Universal count alteration; exact threshold 1 impossible, 2 sufficient

Section 6 - Realizable probe complexity under CNF constraints

Section 7 - Parity as a width-separated example; exact value for ow < r

Section 8 - Open problems and conjectures

Section 9 - Conclusion

---

2. Local Model Statistics in $\#SAT$

Let F be a CNF formula over a finite set of Boolean variables V.

We write

$$\mathrm{Vars}(F) \subseteq V$$

for the set of variables occurring in F.

2.1 Assignments and model counts

An assignment is a mapping

$$\alpha: V \to \{0,1\}.$$

An assignment α satisfies F if $F(\alpha) = 1$. The set of satisfying assignments of F is denoted

$$\mathrm{Mod}(F) = \{\alpha: V \to \{0,1\} \mid F(\alpha) = 1\}.$$

The model count of F is

$$\#F := \mid \mathrm{Mod}(F) \mid.$$

2.2 Local patterns

Fix a tuple of pairwise distinct variables

$$X = (x_1, \dots, x_r)$$

with $x_1, \dots, x_r \in \mathrm{Vars}(F)$.

For an assignment α, define its restriction to X by

$$\alpha \mid_X := (\alpha(x_1), \dots, \alpha(x_r)) \in \{0,1\}^r.$$

We call $\alpha \mid_X$ the **local pattern induced by α on X**.

2.3 Local model distribution

For each pattern $a \in \{0,1\}^r$, define

$$N_F(a) := \mid \{\alpha \in \mathrm{Mod}(F) \mid \alpha \mid_X = a\} \mid.$$

The vector

$$N_F = (N_F(a))_{a \in \{0,1\}^r}$$

is called the **local model distribution of F on X**.

By definition,

$$\sum_{a \in \{0,1\}^r} N_F\,(a) = \#F.$$

2.4 Local model statistics

A **local model statistic of order** r is any function

$$\phi: \mathbb{N}^{2^r} \to \mathbb{R}$$

whose value on F depends only on the local model distribution N_F on some fixed tuple X of length r.

Thus, for a fixed tuple X, every local statistic can be written as a function of the vector N_F.

2.5 Partition statistics

Let

$$\Pi = \{B_1, \dots, B_m\}$$

be a partition of $\{0,1\}^r$.

For each block B_j, define its associated block mass by

$$M_F(B_j) := \sum_{a \in B_j} N_F\,(a).$$

The vector

$$M_F^{\Pi} := (M_F(B_1), \dots, M_F(B_m))$$

is called the **partition statistic induced by** Π.

Since Π is a partition,

$$\sum_{j=1}^{m} M_F\,(B_j) = \#F.$$

2.6 CNF extensions

Let Y be a finite set of fresh variables such that

$$Y \cap \mathrm{Vars}(F) = \emptyset.$$

Let H be a CNF formula over variables contained in

$$\mathrm{Vars}(F) \cup Y.$$

Then

$$F' := F \wedge H$$

is called a **CNF extension** of F.

2.7 Probes

Let $X = (x_1, \dots, x_r)$ be a fixed tuple of variables.

A CNF extension H is called a **probe on** X if there exists a function

$$q_H : \mathbb{N}^{2^r} \to \mathbb{Z}$$

such that for every CNF formula F with $x_1, \dots, x_r \in \mathrm{Vars}(F)$,

$$\#(F \wedge H) - \#F = q_H(N_F).$$

Thus a probe is a selected CNF extension whose count effect depends only on the local model distribution on X, uniformly over all formulas containing X.

We write

$$Q_H(F) := \#(F \wedge H) - \#F$$

for the probe observation induced by H.

2.8 Indicator probes

Let $S \subseteq \{0,1\}^r$.

A probe H_S on X is called an **indicator probe for** S if for every CNF formula F containing X,

$$Q_{H_S}(F) = \sum_{a \in S} N_F\,(a).$$

Equivalently, the probe observation records exactly the number of models of F whose pattern on X lies in S.

Indicator probes will be used in the sequel to recover partition statistics and full local model distributions.

2.9 Notational convention

When $r = 1,2,3$, we will often write

$$N_F(0), N_F(1), N_F(00), \dots, N_F(11), N_F(000), \dots, N_F(111),$$

instead of $N_F(a)$ with $a \in \{0,1\}^r$, whenever no ambiguity can arise.

---

3. Abstract Probe Complexity

In this section we define the abstract probe complexity of a local model statistic and determine it exactly for partition statistics.

Throughout, let

$$X = (x_1, \dots, x_r)$$

be a fixed tuple of pairwise distinct variables, and let F be a CNF formula containing X.

Let

$$N_F = (N_F(a))_{a \in \{0,1\}^r}$$

denote the local model distribution of F on X.

3.1 Probe families

Let

$$\mathcal{H} = \{H_1, \dots, H_k\}$$

be a finite family of probes on X.

For each $i \in \{1, \dots, k\}$, define the corresponding probe observation by

$$Q_{H_i}(F) := \#(F \wedge H_i) - \#F.$$

The associated observation vector is

$$Q_{\mathcal{H}}(F) := (Q_{H_1}(F), \dots, Q_{H_k}(F)).$$

Since each H_i is a probe on X, every component of $Q_{\mathcal{H}}(F)$ depends only on the local model distribution N_F.

3.2 Determination of a statistic

Let ϕ be a local model statistic of order r.

A probe family $\mathcal{H} = \{H_1, \dots, H_k\}$ is said to **determine** ϕ if there exists a function

$$g: \mathbb{Z}^k \times \mathbb{N} \to \mathbb{R}$$

such that for every CNF formula F containing X,

$$\phi(N_F) = g(Q_{\mathcal{H}}(F), \#F).$$

Thus ϕ is recoverable from the probe observations together with the total model count.

3.3 Abstract probe complexity

The **abstract probe complexity** of a local model statistic ϕ, denoted

$$\mathrm{PC}(\phi),$$

is the minimum integer k such that there exists a probe family of size k determining ϕ.

If no such finite family exists, we write

$$\mathrm{PC}(\phi) = \infty.$$

3.4 Indicator probes as abstract observables

Indicator probes yield distinguished scalar observables on the local model distribution.

Let $S \subseteq \{0,1\}^r$, and let H_S be an indicator probe for S. Then

$$Q_{H_S}(F) = \sum_{a \in S} N_F\,(a).$$

In particular, each indicator probe computes a block sum of the local model distribution.

This observation will be sufficient to establish the exact abstract probe complexity of partition statistics.

3.5 Partition statistics

Let

$$\Pi = \{B_1, \dots, B_m\}$$

be a partition of $\{0,1\}^r$.

Recall from Section 2 that the associated partition statistic is

$$M_F^{\Pi} = (M_F(B_1), \dots, M_F(B_m)), M_F(B_j) = \sum_{a \in B_j} N_F\,(a).$$

Since

$$\sum_{j=1}^{m} M_F\,(B_j) = \#F,$$

the vector M_F^{Π} has $m - 1$ independent degrees of freedom once $\#F$ is fixed.

3.6 The Partition Probe Theorem

We now determine the exact abstract probe complexity of every partition statistic.

Theorem 3.1 (Partition Probe Theorem)

Let

$$\Pi = \{B_1, \dots, B_m\}$$

be a partition of $\{0,1\}^r$.

Then the abstract probe complexity of the partition statistic M_F^{Π} is

$$\mathrm{PC}(M_F^{\Pi}) = m - 1.$$

Proof

Upper bound.
For each $j \in \{1, \dots, m-1\}$, let H_{B_j} be an indicator probe for the block B_j. Then, for every CNF formula F containing X,

$$Q_{H_{B_j}}(F) = \sum_{a \in B_j} N_F(a) = M_F(B_j).$$

Hence the probe family

$$\mathcal{H} = \{H_{B_1}, \dots, H_{B_{m-1}}\}$$

determines the first $m-1$ block masses. Since

$$M_F(B_m) = \#F - \sum_{j=1}^{m-1} M_F(B_j),$$

the last block mass is recovered from $\#F$. Therefore

$$\mathrm{PC}(M_F^{\Pi}) \leq m - 1.$$

Lower bound.
The partition statistic M_F^{Π} has $m-1$ independent degrees of freedom once $\#F$ is fixed. Therefore no collection of fewer than $m-1$ scalar probe observations can determine M_F^{Π} for all formulas in general. Hence

$$\mathrm{PC}(M_F^{\Pi}) \geq m - 1.$$

Combining the two bounds yields

$$\mathrm{PC}(M_F^{\Pi}) = m - 1.$$

■

3.7 Immediate consequences

We record three direct consequences that will be used repeatedly in the sequel.

Corollary 3.2

Let $X = (x)$. The partition statistic induced by

$$\{\{0\}, \{1\}\}$$

has abstract probe complexity 1.

Corollary 3.3

Let $X = (x, u)$. The partition statistic induced by

$$\{\{00,11\},\{01,10\}\}$$

has abstract probe complexity 1.

Corollary 3.4

Let $X = (x_1, \dots, x_r)$. The full local model distribution on X has abstract probe complexity

$$2^r - 1.$$

These consequences will be expanded in Section 4.

---

4. Partition Statistics and Local Tomography

In this section we record several direct consequences of Theorem 3.1 and fix the main examples used throughout the paper.

Throughout, let

$$X = (x_1, \dots, x_r)$$

be a fixed tuple of pairwise distinct variables occurring in a CNF formula F, and let

$$N_F = (N_F(a))_{a \in \{0,1\}^r}$$

denote the local model distribution on X.

For brevity, when $r = 1,2,3$, we use the shorthand notation

$$N_F(0), N_F(1), N_F(00), \dots, N_F(11), N_F(000), \dots, N_F(111)$$

instead of $N_F(a)$ with $a \in \{0,1\}^r$.

4.1 Single-variable bias

Let $r = 1$, and let $X = (x)$.

Consider the partition

$$\Pi_x = \{\{0\}, \{1\}\}.$$

Its associated block masses are

$$M_F(\{0\}) = N_F(0), M_F(\{1\}) = N_F(1).$$

Since

$$N_F(0) + N_F(1) = \#F,$$

Theorem 3.1 yields the following.

Proposition 4.1

The partition statistic induced by Π_x has abstract probe complexity

$$\mathrm{PC}(\Pi_x) = 1.$$

In particular, the local bias of a variable can be determined with one probe.

4.2 Equality and disequality on two variables

Let $r = 2$, and let $X = (x, u)$.

Consider the partition

$$\Pi_= = \{\{00,11\}, \{01,10\}\}.$$

Its block masses are

$$M_F(\{00,11\}) = N_F(00) + N_F(11),$$
$$M_F(\{01,10\}) = N_F(01) + N_F(10).$$

The first block counts the models satisfying $x = u$, and the second counts the models satisfying $x \neq u$.

Proposition 4.2

The partition statistic induced by $\Pi_=$ has abstract probe complexity

$$\mathrm{PC}(\Pi_=) = 1.$$

In particular, the equality/disequality statistic of a pair of variables can be determined with one probe.

4.3 Parity on r variables

Let $r \geq 2$, and let

$$X = (x_1, \dots, x_r).$$

Define

$$E_r := \{a \in \{0,1\}^r : a_1 \oplus \cdots \oplus a_r = 0\},$$
$$O_r := \{a \in \{0,1\}^r : a_1 \oplus \cdots \oplus a_r = 1\}.$$

These sets form a partition

$$\Pi_{\oplus} = \{E_r, O_r\}.$$

The corresponding block masses are

$$M_F(E_r) = \sum_{a \in E_r} N_F(a), M_F(O_r) = \sum_{a \in O_r} N_F(a).$$

Thus $M_F(O_r)$ is the number of models of F whose pattern on X has odd parity.

Proposition 4.3

The partition statistic induced by $\Pi_{\oplus}$ has abstract probe complexity

$$\mathrm{PC}(\Pi_{\oplus}) = 1.$$

In particular, the parity statistic on r variables can be determined with one probe.

4.4 Complete local model tomography

Consider the discrete partition

$$\Pi_{\mathrm{disc}} = \{\{a\} : a \in \{0,1\}^r\}.$$

This partition has 2^r singleton blocks, and its associated statistic is exactly the full local model distribution

$$N_F = (N_F(a))_{a \in \{0,1\}^r}.$$

Applying Theorem 3.1 immediately yields:

Theorem 4.4

The full local model distribution on r variables has abstract probe complexity

$$\mathrm{PC}(N_F) = 2^r - 1.$$

Equivalently, complete local model tomography on r variables requires exactly $2^r - 1$ probes.

4.5 Order-d local tomography

Let k be the size of a fixed ambient set of original variables, and let $1 \leq d \leq k$.

For each $r \in \{1, \dots, d\}$, there are

$$\binom{k}{r}$$

choices of r-element subsets of variables. For each such subset, complete local tomography requires exactly $2^r - 1$ probes by Theorem 4.4.

Therefore:

Corollary 4.5

The total abstract probe complexity required to reconstruct all complete local marginals up to order d is

$$\sum_{r=1}^{d} \binom{k}{r} (2^r - 1).$$

4.6 Partition hierarchies

Theorem 3.1 also yields a general formula for arbitrary families of partition statistics.

For each $r \in \{1, \dots, d\}$, let

$$\Pi_r$$

be a partition of $\{0,1\}^r$ into $|\Pi_r|$ blocks. Suppose that for every r-element subset of variables one wishes to reconstruct the partition statistic induced by Π_r.

Then Theorem 3.1 gives:

Corollary 4.6

The total abstract probe complexity required to reconstruct all partition statistics up to order d is

$$\sum_{r=1}^{d} \binom{k}{r} (|\Pi_r| - 1).$$

4.7 Interpretation

The preceding results show that the abstract probing cost of a local statistic is determined by the number of independent block masses required to specify it.

In particular:

- coarse statistics such as bias, equality/disequality, and parity have constant abstract probe complexity;
- complete local tomography has abstract probe complexity exponential in the order r.

This distinction will be used in the sequel to separate the intrinsic information dimension of a statistic from the structural cost of realizing it through CNF probes.

---

5. Universal Count Alteration

In Sections 3 and 4 the objective was the exact determination of local model statistics. In the present section we study a different task: not reconstruction, but the existence of a probe that guarantees a strict increase of the model count for every satisfiable formula.

Throughout this section, let F be a satisfiable CNF formula.

5.1 Universally count-altering families

Let

$$X = (x_1, \dots, x_r)$$

be a fixed tuple of pairwise distinct variables occurring in F.

A finite family of probes

$$\mathcal{H} = \{H_1, \dots, H_t\}$$

on X is called **universally count-altering** if, for every satisfiable CNF formula F containing X, there exists $i \in \{1, \dots, t\}$ such that

$$\#(F \wedge H_i) > \#F.$$

The quantity studied in this section is the minimum size of such a family. This is distinct from abstract probe complexity, since the present task concerns universal strict count increase rather than reconstruction of a statistic.

5.2 The one-probe impossibility

We first show that no single non-tautological probe can be universally count-altering.

Let $G(X, y)$ be a probe using one fresh variable y, where

$$X = (x_1, \dots, x_r).$$

For each pattern $a \in \{0,1\}^r$, define

$$t_G(a) := |\{b \in \{0,1\} : G(a, b) = 1\}|.$$

Thus $t_G(a) \in \{0,1,2\}$, and for every formula F,

$$\#(F \wedge G) = \sum_{a \in \{0,1\}^r} N_F(a)\, t_G(a).$$

Proposition 5.1

Let $r \geq 1$. No single non-tautological probe is universally count-altering.

Proof

Assume that G is universally count-altering.

Fix an arbitrary pattern $a \in \{0,1\}^r$, and let F_a be a satisfiable CNF formula whose models realize only the pattern a on X. Then $\#F_a > 0$, and

$$\#(F_a \wedge G) = \#F_a \cdot t_G(a).$$

Since G is universally count-altering, we must have

$$\#(F_a \wedge G) > \#F_a,$$

hence $t_G(a) > 1$. Because $t_G(a) \in \{0,1,2\}$, it follows that $t_G(a) = 2$.

Since a was arbitrary,

$$t_G(a) = 2 \text{ for all } a \in \{0,1\}^r.$$

Therefore $G(a,b) = 1$ for both values of b, for every pattern a. Hence G is satisfied by every assignment to (X,y), i.e. G is tautological. This contradicts the assumption that G is non-tautological. ■

5.3 The case of two variables

We now show that for $r = 2$, two probes suffice.

Let

$$X = (x,u).$$

Define the probes

$$H_=(x,u,y) := (x \vee \neg u \vee y) \wedge (\neg x \vee u \vee y),$$
$$H_\neq(x,u,y) := (x \vee u \vee y) \wedge (\neg x \vee \neg u \vee y).$$

For each pattern $(a,b) \in \{0,1\}^2$, direct inspection shows:

- $H_=$ leaves y free exactly on 00 and 11;
- $H_\neq$ leaves y free exactly on 01 and 10.

Accordingly,

$$\#(F \wedge H_=) = \#F + (N_F(00) + N_F(11)),$$
$$\#(F \wedge H_\neq) = \#F + (N_F(01) + N_F(10)).$$

Since

$$N_F(00) + N_F(01) + N_F(10) + N_F(11) = \#F > 0,$$

at least one of the two added terms is positive.

Theorem 5.2

For $r = 2$, the family $\{H_=, H_\neq\}$ is universally count-altering.

Proof

For every satisfiable CNF formula F containing x and u,

$$\#(F \wedge H_{=}) - \#F = N_F(00) + N_F(11),$$
$$\#(F \wedge H_{\neq}) - \#F = N_F(01) + N_F(10).$$

Since the sum of these two quantities is $\#F > 0$, at least one of them is positive. Therefore at least one of the two probes strictly increases the model count. ■

Corollary 5.3

For $r = 2$, the minimum size of a universally count-altering family is exactly 2.

Proof

The lower bound follows from Proposition 5.1. The upper bound follows from Theorem 5.2. ■

5.4 The case of three variables

Let

$$X = (x, u, v).$$

Define

$$E_3 := \{a \in \{0,1\}^3 : a_1 \oplus a_2 \oplus a_3 = 0\}, O_3 := \{a \in \{0,1\}^3 : a_1 \oplus a_2 \oplus a_3 = 1\}.$$

Let C_a denote the clause falsified exactly by the pattern $a \in \{0,1\}^3$. Define

$$H_{\text{even}}(x, u, v, y) := \bigwedge_{a \in O_3} (C_a \vee y),$$
$$H_{\text{odd}}(x, u, v, y) := \bigwedge_{a \in E_3} (C_a \vee y).$$

Then:

- H_{even} leaves $y$free exactly on the even-parity patterns;
- H_{odd} leaves $y$free exactly on the odd-parity patterns.

Hence

$$\#(F \wedge H_{\text{even}}) = \#F + \sum_{a \in E_3} N_F(a),$$
$$\#(F \wedge H_{\text{odd}}) = \#F + \sum_{a \in O_3} N_F(a).$$

Since

$$\sum_{a \in E_3} N_F(a) + \sum_{a \in O_3} N_F(a) = \#F > 0,$$

at least one of the two summands is positive.

Theorem 5.4

For $r = 3$, the family $\{H_{\text{even}}, H_{\text{odd}}\}$is universally count-altering.

Corollary 5.5

For $r = 3$, the minimum size of a universally count-altering family is exactly 2.

Proof

The lower bound follows from Proposition 5.1, and the upper bound from Theorem 5.4. ■

5.5 The general case

The preceding construction extends immediately to all arities $r \geq 2$.

Let

$$X = (x_1, \dots, x_r),$$

and define

$$E_r := \{a \in \{0,1\}^r : a_1 \oplus \cdots \oplus a_r = 0\}, O_r := \{a \in \{0,1\}^r : a_1 \oplus \cdots \oplus a_r = 1\}.$$

Let C_a denote the clause falsified exactly by the pattern $a \in \{0,1\}^r$. Define

$$H_{\text{even}}^{(r)} := \bigwedge_{a \in O_r} (C_a \vee y),$$

$$H_{\text{odd}}^{(r)} := \bigwedge_{a \in E_r} (C_a \vee y).$$

Then

$$\#(F \wedge H_{\text{even}}^{(r)}) = \#F + \sum_{a \in E_r} N_F(a),$$
$$\#(F \wedge H_{\text{odd}}^{(r)}) = \#F + \sum_{a \in O_r} N_F(a).$$

Since

$$\sum_{a \in E_r} N_F(a) + \sum_{a \in O_r} N_F(a) = \#F > 0,$$

at least one of the two summands is positive.

Theorem 5.6

Let $r \geq 2$. The family

$$\left\{H_{\text{even}}^{(r)}, H_{\text{odd}}^{(r)}\right\}$$

is universally count-altering.

Corollary 5.7

Let $r \geq 2$. The minimum size of a universally count-altering family on $r$variables is exactly 2.

Proof

The lower bound follows from Proposition 5.1. The upper bound follows from Theorem 5.6. ■

5.6 Comparison with tomography

The preceding results establish a sharp distinction between exact reconstruction and universal count alteration.

On one hand, complete local tomography on r variables requires exactly

$$2^r - 1$$

probes by Theorem 4.4.

On the other hand, universal strict increase of the model count requires exactly

$$2$$

probes for every $r \geq 2$ by Corollary 5.7.

Thus the cost of forcing a guaranteed count change is constant, whereas the cost of complete local reconstruction is exponential in the arity. This separation will be contrasted in the next sections with the effect of structural CNF constraints on probe realizability.

---

6. Realizable Probe Complexity under CNF Constraints

Sections 3–5 concern abstract probing: they quantify the minimum number of probes required to determine a local statistic or to guarantee universal count alteration, without imposing syntactic restrictions on the probes beyond the requirement that they be CNF extensions. In this section we introduce a structural refinement in which probes must be realized by explicit CNF formulas subject to additional constraints.

Throughout, let F be a CNF formula over a finite set of original variables V, and let

$$X = (x_1, \dots, x_r)$$

be a fixed tuple of pairwise distinct variables from V.

Let $Y \cap V = \emptyset$ be a finite set of fresh variables.

6.1 Realizable probes

A **realizable probe on** X is a CNF formula

$$H(X, Z, Y),$$

where $Z \subseteq V \setminus \{x_1, \dots, x_r\}$ is a set of additional original variables and Y is a set of fresh variables, such that the induced count difference

$$\#(F \wedge H) - \#F$$

depends only on the local model distribution of F on X, uniformly over all CNF formulas F containing X.

Thus a realizable probe may use original variables outside X, but its count effect must still be determined solely by the local statistic on X.

Every realizable probe therefore induces an abstract probe in the sense of Section 2.

6.2 Structural classes of realizable probes

We now introduce several structural restrictions on realizable probes.

Definition 6.1 (Connected probe)

A realizable probe H is **connected** if, in the variable–clause incidence graph of $F \wedge H$, every fresh variable occurring in H lies in the same connected component as at least one original variable of F.

Definition 6.2 (Non-tautological probe)

A realizable probe H is **non-tautological** if no clause of H is tautological.

Definition 6.3 (Old-width of a clause)

Let C be a clause occurring in a realizable probe H. The **old-width** of C is the number of original variables from V occurring in C.

Definition 6.4 (Old-width of a probe)

The **old-width** of a realizable probe H, denoted

$$\mathrm{ow}(H),$$

is the maximum old-width of its clauses.

Definition 6.5 (Width-bounded realizability class)

For $w \geq 1$, let

$$\mathcal{C}_{\mathrm{ow} \leq w}$$

denote the class of realizable probes Hsatisfying

$$\mathrm{ow}(H) \leq w.$$

Similarly, we write

$$\mathcal{C}_{\mathrm{ow} < w}$$

for the class of realizable probes H satisfying

$$\mathrm{ow}(H) < w.$$

6.3 Realizable probe complexity

Let ϕ be a local model statistic on X, and let $\mathcal{C}$ be a class of realizable probes.

The **realizable probe complexity** of ϕ in $\mathcal{C}$, denoted

$$\mathrm{RPC}_{\mathcal{C}}(\phi),$$

is the minimum size of a probe family

$$\mathcal{H} = \{H_1, \dots, H_t\} \subseteq \mathcal{C}$$

that determines ϕ.

If no such finite family exists, we write

$$\mathrm{RPC}_{\mathcal{C}}(\phi) = \infty.$$

6.4 Abstract versus realizable probing

Every realizable probe family induces, by forgetting its explicit CNF implementation, an abstract probe family in the sense of Section 3. Consequently, realizability constraints can only increase probing cost.

Proposition 6.6

Let ϕ be a local model statistic and let $\mathcal{C}$ be any class of realizable probes. Then

$$\mathrm{RPC}_{\mathcal{C}}(\phi) \geq \mathrm{PC}(\phi).$$

Proof

Any probe family in $\mathcal{C}$ determining ϕ is, by definition, a family of realizable probes whose count effects depend only on the local model distribution on X. Hence, after forgetting their CNF implementation, they form an abstract probe family determining ϕ. Therefore the minimum size over $\mathcal{C}$ cannot be smaller than the unconstrained minimum $\mathrm{PC}(\phi)$. ■

Definition 6.7 (Realizability gap)

Let ϕ be a local model statistic and let $\mathcal{C}$ be a class of realizable probes. We say that ϕ exhibits a **realizability gap** in $\mathcal{C}$ if

$$\mathrm{RPC}_{\mathcal{C}}(\phi) > \mathrm{PC}(\phi).$$

6.5 Universally count-altering families in the realizable setting

The universally count-altering families constructed in Section 5 are explicitly realizable by CNF probes. In particular, for every $r \geq 2$, the parity probes

$$H_{\mathrm{even}}^{(r)} = \bigwedge_{a \in O_r} (C_a \vee y), H_{\mathrm{odd}}^{(r)} = \bigwedge_{a \in E_r} (C_a \vee y)$$

are realizable, non-tautological, and connected.

Indeed, every clause of these probes contains both the fresh variable $y$and original variables from X, so in the variable–clause incidence graph the vertex y lies in the same connected component as X.

Therefore the exact threshold of Section 5 remains valid in the realizable setting, provided no old-width restriction is imposed.

Proposition 6.8

Let $r \geq 2$. In the class of connected non-tautological realizable probes, the minimum size of a universally count-altering family on r variables is exactly 2.

Proof

The lower bound follows from Proposition 5.1, which is purely semantic. The upper bound is realized by the pair $H_{\mathrm{even}}^{(r)}, H_{\mathrm{odd}}^{(r)}$, which are explicit connected non-tautological realizable probes. ■

6.6 Parity as a local statistic

Fix $r \geq 2$, and let

$$\phi_{\oplus,r}$$

denote the odd-parity statistic on $X = (x_1, \dots, x_r)$, i.e.

$$\phi_{\oplus,r}(F) = \sum_{a \in O_r} N_F\,(a),$$

where

$$O_r = \{a \in \{0,1\}^r : a_1 \oplus \cdots \oplus a_r = 1\}.$$

By Proposition 4.3,

$$\mathrm{PC}(\phi_{\oplus,r}) = 1.$$

We now show that this equality fails under old-width restrictions.

6.7 Exact parity probes

We first isolate the class of probes relevant to the lower bound.

Definition 6.9 (Exact parity probe)

A realizable probe $H(X, y)$ with one fresh variable $y$is an **exact parity probe** on $X = (x_1, \dots, x_r)$ if the following hold for every pattern $a \in \{0,1\}^r$:

1. if $a \in E_r$, then both assignments $y = 0$ and $y = 1$ satisfy $H(a, y)$;
2. if $a \in O_r$, then exactly one value of y satisfies $H(a, y)$.

In particular, every exact parity probe satisfies

$$\#(F \wedge H) - \#F = \phi_{\oplus,r}(F)$$

for every CNF formula F containing X.

6.8 A clause-level parity obstruction

We now prove a structural lemma showing that clauses of old-width strictly smaller than r cannot separate parity exactly.

Lemma 6.10

Let C be a clause on $x_1, \dots, x_r$ whose old-width is strictly smaller than r. If an assignment $a \in \{0,1\}^r$ falsifies C, then there exists an assignment $b \in \{0,1\}^r$ of opposite parity that also falsifies C.

Proof

Since the old-width of C is $<r$, at least one variable x_i does not occur in C. Let b be obtained from a by flipping the value of x_i. Because x_i does not occur in C, all literals of C have the same truth values under a and b. Hence b also falsifies C. Since a and b differ in exactly one bit, they have opposite parity. ■

6.9 Width barrier for a single exact parity probe

We now derive the first strict realizability gap.

Theorem 6.11

Let $r \geq 2$. No exact parity probe on $r$variables exists in the class

$$\mathcal{C}_{\mathrm{ow}<r}.$$

Equivalently,

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}<r}}(\phi_{\oplus,r}) > 1.$$

Proof

Assume that $H(X, y)$ is an exact parity probe on $X = (x_1, \dots, x_r)$ and that every clause of H has old-width $<r$.

Substitute $y = 0$ into H, and let

$$H_0(X) := H(X, 0).$$

By Definition 6.9, H_0 is satisfied exactly by the even-parity patterns on X: if a pattern is in E_r, then $y = 0$ is allowed; if it is in O_r, then $y = 0$ is not allowed.

Now let $a \in O_r$. Since $H_0(a) = 0$, some clause C of H_0 is falsified by a. By Lemma 6.10, there exists $b \in E_r$ that also falsifies C. Hence $H_0(b) = 0$, contradicting the fact that H_0 must be satisfied by all even-parity patterns.

Therefore no such exact parity probe exists. ■

6.10 First realizability gap

Theorem 6.11 yields the first strict separation between abstract and realizable probing.

Corollary 6.12

For parity on r variables,

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}<r}}(\phi_{\oplus,r}) > \mathrm{PC}(\phi_{\oplus,r}) = 1.$$

Thus parity exhibits a realizability gap under the old-width restriction $\mathrm{ow} < r$.

The next section sharpens this result by proving that, although one exact parity probe of old-width $<r$ is impossible, two such probes always suffice.

---

7. Parity as a Width-Separated Example

In this section we determine the exact realizable probe complexity of parity under the old-width restriction $\mathrm{ow} < r$.

Throughout, let

$$X = (x_1, \dots, x_r)$$

be a fixed tuple of pairwise distinct variables with $r \geq 2$, and let

$$\phi_{\oplus,r}(F) := \sum_{a \in O_r} N_F\,(a)$$

denote the odd-parity statistic on X, where

$$E_r := \{a \in \{0,1\}^r : a_1 \oplus \cdots \oplus a_r = 0\}, O_r := \{a \in \{0,1\}^r : a_1 \oplus \cdots \oplus a_r = 1\}.$$

By Proposition 4.3,

$$\mathrm{PC}(\phi_{\oplus,r}) = 1.$$

By Theorem 6.11,

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}<r}}(\phi_{\oplus,r}) > 1.$$

We now prove the matching upper bound.

7.1 Decomposition of odd parity

Fix the first variable x_1. Define the two disjoint sets

$$S_0 := \{a \in \{0,1\}^r : a_1 = 0,\ a_2 \oplus \cdots \oplus a_r = 1\},$$
$$S_1 := \{a \in \{0,1\}^r : a_1 = 1,\ a_2 \oplus \cdots \oplus a_r = 0\}.$$

Then

$$O_r = S_0 \cup S_1, S_0 \cap S_1 = \emptyset.$$

Hence, for every formula F,

$$\phi_{\oplus,r}(F) = M_F(S_0) + M_F(S_1).$$

It therefore suffices to realize one probe for S_0 and one probe for S_1.

7.2 Auxiliary parity clause sets on $r-1$ variables

Let

$$X' = (x_2, \dots, x_r).$$

For each pattern $b \in \{0,1\}^{r-1}$, let $C_b(X')$ denote the clause on X' falsified exactly by b, i.e.

$$C_b(X') = \bigvee_{i=2}^{r} \ell_i,$$

where

$$\ell_i = \begin{cases} x_i & \text{if } b_{i-1} = 0, \\ \neg x_i & \text{if } b_{i-1} = 1. \end{cases}$$

Define the clause sets

$$\mathcal{P}_{r-1}^{\text{even}} := \{\, C_b(X') : b \in O_{r-1} \,\},$$
$$\mathcal{P}_{r-1}^{\text{odd}} := \{\, C_b(X') : b \in E_{r-1} \,\},$$

where

$$E_{r-1} := \{b \in \{0,1\}^{r-1} : b_1 \oplus \cdots \oplus b_{r-1} = 0\}, O_{r-1} := \{b \in \{0,1\}^{r-1} : b_1 \oplus \cdots \oplus b_{r-1} = 1\}.$$

Let

$$P_{r-1}^{\text{even}}(X') := \bigwedge_{C \in \mathcal{P}_{r-1}^{\text{even}}} C,$$

$$P_{r-1}^{\mathrm{odd}}(X') := \bigwedge_{C \in \mathcal{P}_{r-1}^{\mathrm{odd}}} C.$$

By construction:

- P_{r-1}^{even} is satisfied exactly by the even-parity patterns on X';
- P_{r-1}^{odd} is satisfied exactly by the odd-parity patterns on X'.

Moreover, every clause in $\mathcal{P}_{r-1}^{\mathrm{even}} \cup \mathcal{P}_{r-1}^{\mathrm{odd}}$ has old-width $r-1$.

7.3 Two realizable parity probes

Let y_0 and y_1 be fresh variables. Define

$$H_0(X, y_0) := (x_1 \vee y_0) \ \wedge \bigwedge_{C \in \mathcal{P}_{r-1}^{\mathrm{odd}}} (C \vee y_0),$$

$$H_1(X, y_1) := (\neg x_1 \vee y_1) \ \wedge \bigwedge_{C \in \mathcal{P}_{r-1}^{\mathrm{even}}} (C \vee y_1).$$

The first probe is intended to detect S_0, and the second S_1.

Lemma 7.1

For every pattern $a \in \{0,1\}^r$:

1. $H_0(a, y_0)$ leaves y_0 free if and only if $a \in S_0$.
2. $H_1(a, y_1)$ leaves y_1 free if and only if $a \in S_1$.

Proof

We prove (1). Let $a = (a_1, \dots, a_r)$.

The clause $(x_1 \vee y_0)$ is satisfied independently of y_0 exactly when $a_1 = 0$. If $a_1 = 1$, it reduces to y_0, and therefore forces $y_0 = 1$.

Now consider the clauses $(C \vee y_0)$ with $C \in \mathcal{P}_{r-1}^{\mathrm{odd}}$. By definition of P_{r-1}^{odd}, all such clauses are satisfied independently of y_0 exactly when $(a_2, \dots, a_r)$ has odd parity. Otherwise at least one such clause reduces to y_0, and therefore forces $y_0 = 1$.

Hence y_0 is free exactly when both conditions hold:

$$a_1 = 0 \text{and} a_2 \oplus \cdots \oplus a_r = 1,$$

that is, exactly when $a \in S_0$.

The proof of (2) is analogous. The clause $(\neg x_1 \vee y_1)$ leaves y_1 free exactly when $a_1 = 1$, and the clauses in $\mathcal{P}_{r-1}^{\text{even}}$ leave y_1 free exactly when $(a_2, \dots, a_r)$ has even parity. Thus y_1 is free exactly on S_1. ■

7.4 Reconstruction of parity with two probes

For $i \in \{0,1\}$, define

$$Q_{H_i}(F) := \#(F \wedge H_i) - \#F.$$

Proposition 7.2

The probes $H_0$and H_1satisfy

$$Q_{H_0}(F) = M_F(S_0), Q_{H_1}(F) = M_F(S_1).$$

Consequently,

$$\phi_{\oplus,r}(F) = Q_{H_0}(F) + Q_{H_1}(F).$$

Proof

By Lemma 7.1, each satisfying assignment $\alpha \in \mathrm{Mod}(F)$ contributes:

- two extensions to $F \wedge H_0$if $\alpha \mid_X \in S_0$,
- one extension otherwise.

Therefore

$$\#(F \wedge H_0) = \#F + M_F(S_0),$$

and hence

$$Q_{H_0}(F) = M_F(S_0).$$

Similarly,

$$Q_{H_1}(F) = M_F(S_1).$$

Since $O_r = S_0 \cup S_1$ is a disjoint union,

$$\phi_{\oplus,r}(F) = M_F(O_r) = M_F(S_0) + M_F(S_1) = Q_{H_0}(F) + Q_{H_1}(F).$$

■

7.5 Width bound

We now verify that both probes belong to the class $\mathcal{C}_{\mathrm{ow}<r}$.

Lemma 7.3

Every clause of H_0 and H_1 has old-width at most $r - 1$.

Proof

The clauses $(x_1 \vee y_0)$ and $(\neg x_1 \vee y_1)$ have old-width 1.

Each clause in $\mathcal{P}_{r-1}^{\mathrm{odd}} \cup \mathcal{P}_{r-1}^{\mathrm{even}}$ is a clause on the variables $x_2, \dots, x_r$, hence has old-width exactly $r - 1$. Appending y_0 or y_1 does not change old-width. Therefore every clause of H_0 and H_1 has old-width at most $r - 1 < r$. ■

7.6 Exact realizable probe complexity under $\mathrm{ow} < r$

We can now combine Theorem 6.11 with Proposition 7.2 and Lemma 7.3.

Theorem 7.4

Let $r \geq 2$. Then

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}<r}}(\phi_{\oplus,r}) = 2.$$

Proof

By Theorem 6.11,

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}<r}}(\phi_{\oplus,r}) > 1.$$

By Proposition 7.2, the family $\{H_0, H_1\}$ determines $\phi_{\oplus,r}$. By Lemma 7.3, both H_0 and H_1 belong to $\mathcal{C}_{\mathrm{ow}<r}$. Hence

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}<r}}(\phi_{\oplus,r}) \leq 2.$$

Therefore

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}<r}}(\phi_{\oplus,r}) = 2.$$

■

7.7 Exact abstract/realizable separation for parity

Theorem 7.4 yields the first exact separation between abstract and realizable probe complexity.

Corollary 7.5

Let $r \geq 2$. Then

$$\mathrm{PC}(\phi_{\oplus,r}) = 1 \text{and} \mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}<r}}(\phi_{\oplus,r}) = 2.$$

In particular, parity exhibits a strict realizability gap under the old-width restriction $\mathrm{ow} < r$.

7.8 A general upper bound for smaller old-width

The preceding construction extends to stronger old-width restrictions.

Let $1 \leq w \leq r$. Fix $r - w$ variables among $x_1, \ldots, x_r$, and enumerate all 2^{r-w} assignments to them. For each such restriction σ, the parity condition on the full r-tuple reduces either to even parity or to odd parity on the remaining w variables. Applying the construction of Sections 6.7 and 7.3 to the residual w-variable parity instance under the restriction σ yields one probe for each σ.

This gives the following upper bound.

Proposition 7.6

Let $1 \leq w \leq r$. Then

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}\leq w}}(\phi_{\oplus,r}) \leq 2^{r-w}.$$

Proof

Choose $r - w$ distinguished variables from X. For each assignment $\sigma \in \{0,1\}^{r-w}$ to these variables, the full parity condition on X restricts to either even parity or odd parity on the remaining w variables. By the parity construction on w variables, one can realize an exact probe for the

corresponding residual parity class using clauses of old-width at most w. Repeating this for all 2^{r-w} assignments σ yields a family of size 2^{r-w} that determines $\phi_{\oplus,r}$. ■

7.9 Open quantitative pattern

Proposition 7.6 suggests the following general law.

Conjecture 7.7

For every $r \geq 2$ and every $1 \leq w \leq r$,

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}\leq w}}(\phi_{\oplus,r}) = 2^{r-w}.$$

The cases $w = r$ and $w = r - 1$ are established by Proposition 4.3 and Theorem 7.4, respectively. The first unresolved case is $w = r - 2$, where the conjecture predicts

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}\leq r-2}}(\phi_{\oplus,r}) = 4.$$

This and related questions are recorded in the next section.

---

8. Open Problems and Conjectures

The preceding sections establish an exact abstract theory of partition probing and a first exact separation between abstract and realizable probing for parity under old-width restrictions. Several structural problems remain open.

8.1 General old-width complexity of parity

Section 7 proves that

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}<r}}(\phi_{\oplus,r}) = 2,$$

and more generally gives the upper bound

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow}\leq w}}(\phi_{\oplus,r}) \leq 2^{r-w}.$$

The natural quantitative problem is whether this upper bound is optimal.

Conjecture 8.1

For every $r \geq 2$ and every $1 \leq w \leq r$,

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow} \leq w}}(\phi_{\oplus,r}) = 2^{r-w}.$$

The cases $w = r$ and $w = r - 1$ are established by Proposition 4.3 and Theorem 7.4, respectively.

8.2 The first open case: $w = r - 2$

The first unresolved instance of Conjecture 8.1 is

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow} \leq r-2}}(\phi_{\oplus,r}).$$

By Proposition 7.6,

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow} \leq r-2}}(\phi_{\oplus,r}) \leq 4.$$

The conjectured exact value is therefore the following.

Conjecture 8.2

For every $r \geq 3$,

$$\mathrm{RPC}_{\mathcal{C}_{\mathrm{ow} \leq r-2}}(\phi_{\oplus,r}) = 4.$$

A proof of this case would provide the first nontrivial step beyond the exact threshold $\mathrm{ow} < r$, and would substantially strengthen the evidence for Conjecture 8.1.

8.3 Structural gaps beyond parity

Parity provides the first explicit example of a strict gap between abstract and realizable probe complexity:

$$\mathrm{PC}(\phi_{\oplus,r}) = 1 \mathrm{but} \mathrm{RPC}_{\mathcal{C}_{\mathrm{ow} < r}}(\phi_{\oplus,r}) = 2.$$

This raises the more general question of when such gaps must occur.

Let ϕ be a local model statistic on r variables. One may ask for structural criteria ensuring either

$$\mathrm{RPC}_{\mathcal{C}}(\phi) = \mathrm{PC}(\phi),$$

or a strict inequality

$$\mathrm{RPC}_{\mathcal{C}}(\phi) > \mathrm{PC}(\phi),$$

for natural classes $\mathcal{C}$ of realizable probes.

Problem 8.3

Characterize those local model statistics ϕ for which realizability gaps occur under natural CNF constraints, including old-width bounds, connectedness requirements, and non-separability conditions.

8.4 Realizable probing for general partition statistics

Theorem 3.1 gives the exact abstract probe complexity of every partition statistic:

$$\mathrm{PC}(M_F^{\Pi}) = \mid \Pi \mid -1.$$

In contrast, essentially nothing is known in general about the corresponding realizable probe complexity under structural restrictions.

This leads to the following problem.

Problem 8.4

Given a partition

$$\Pi = \{B_1, \dots, B_m\}$$

of $\{0,1\}^r$ and a realizability class $\mathcal{C}$, determine

$$\mathrm{RPC}_{\mathcal{C}}(M_F^{\Pi}).$$

Even for partitions of low combinatorial complexity, such as parity partitions or Hamming-weight partitions, the exact realizable cost may differ substantially from the abstract value $\mid \Pi \mid -1$.

8.5 Order-resolved probing hierarchies

For each order r, the abstract cost of a partition statistic depends only on the number of its blocks. By contrast, the realizable cost may depend both on the structural complexity of those blocks and on the syntactic restrictions imposed on the probes.

This motivates the study of order-resolved probing hierarchies.

Problem 8.5

For a fixed realizability class $\mathcal{C}$, determine the growth of realizable probe complexity for natural hierarchies of local statistics, including:

1. parity statistics of increasing order;
2. Hamming-weight partition statistics;
3. equality and correlation statistics on tuples of variables;
4. full local model tomography.

In particular, determine whether there exist natural families of local statistics with bounded abstract probe complexity but unbounded realizable probe complexity.

8.6 Perspective

The results of this paper establish two complementary principles.

First, local model statistics in $\#SAT$ admit an exact abstract probing theory: for partition statistics, the probing cost is precisely the number of independent block masses.

Second, structural CNF restrictions can strictly increase this cost, as already witnessed by parity under old-width constraints.

The main open direction is therefore to understand the relation between:

- the information dimension of a local statistic;
- the syntactic restrictions imposed on the probe class;
- the resulting realizable probing cost.

This appears to be the central structural problem suggested by the present work.

---

9. Conclusion

We introduced a probe-based framework for local model statistics in $\#SAT$. For a fixed tuple of variables, every model of a CNF formula induces a local pattern, and the resulting local model distribution gives rise to a natural family of statistics. Within this setting, we defined abstract probe complexity and proved an exact characterization for partition statistics: if a statistic is induced by a partition of the pattern space into $m$blocks, then its abstract probe complexity is exactly $m - 1$.

This characterization yields, as immediate consequences, exact bounds for several natural examples, including single-variable bias, equality and disequality statistics, parity, and complete local model tomography. In particular, complete tomography on $r$variables requires exactly $2^r - 1$ probes, while much coarser statistics can be recovered with constant probing cost.

We then considered a different task, namely universal count alteration: the minimum number of probes required to guarantee a strict increase of the model count for every satisfiable formula. Here

we proved an exact threshold: one non-tautological probe never suffices, while two probes are sufficient for every arity $r \geq 2$.

The second main theme of the paper is the distinction between abstract and realizable probing. Abstract probe complexity measures the information-theoretic cost of recovering a statistic, whereas realizable probe complexity imposes structural CNF constraints on the probes themselves. For parity, this distinction already yields a strict gap: although parity has abstract probe complexity 1, no single exact parity probe exists under the old-width restriction $\mathrm{ow} < r$, while two probes suffice. This provides a first explicit example in which structural constraints strictly increase the probing cost.

Taken together, these results suggest that local probing in $\#SAT$ admits two complementary theories. The first is abstract and information-theoretic, governed by the number of independent local degrees of freedom. The second is structural, governed by the realizability of probes under syntactic CNF constraints. Understanding the relation between these two levels, and in particular the size and nature of the resulting realizability gaps, appears to be the central problem opened by the present work.

References

Adnan Darwiche and Pierre Marquis, *A Knowledge Compilation Map.* Journal of Artificial Intelligence Research, 17:229–264, 2002.

Adnan Darwiche and Pierre Marquis, *Knowledge Compilation.* Annals of Mathematics and Artificial Intelligence, 92(5):1007–1011, 2024.

Adnan Darwiche, *Searching While Keeping a Trace: From Knowledge Compilation to Assumption-Based Reasoning.* In Proceedings of the 21st International Conference on Automated Deduction (CADE 2007), 2007.

Armin Biere, Marijn Heule, Hans van Maaren, and Toby Walsh, editors, *Handbook of Satisfiability. Second Edition.* Frontiers in Artificial Intelligence and Applications 336, IOS Press, 2021.

Benjamin Bergougnoux, Florent Capelli, and Mamadou Moustapha Kanté, *Counting Minimal Transversals of β-Acyclic Hypergraphs.* Journal of Computer and System Sciences, 101:21–30, 2019.

Florent Capelli, Arnaud Durand, and Stefan Mengel. *Hypergraph Acyclicity and Propositional Model Counting.* In Proceedings of the 17th International Conference on Theory and Applications of Satisfiability Testing (SAT 2014), pages 399–414, 2014.

Florent Capelli, *Understanding the Complexity of #SAT Using Knowledge Compilation.* In Proceedings of the 32nd Annual ACM/IEEE Symposium on Logic in Computer Science (LICS 2017), 2017.

Jinbo Huang and Adnan Darwiche, *DPLL with a Trace: From SAT to Knowledge Compilation.* In Proceedings of the Nineteenth International Joint Conference on Artificial Intelligence (IJCAI 2005), pages 156–162, 2005.

Kuldeep S. Meel and Ofer Strichman, editors, *25th International Conference on Theory and Applications of Satisfiability Testing (SAT 2022).* LIPIcs 236, Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2022.

Umut Oztok and Adnan Darwiche, *An Exhaustive DPLL Algorithm for Model Counting.* Journal of Artificial Intelligence Research, 62:1–32, 2018.